

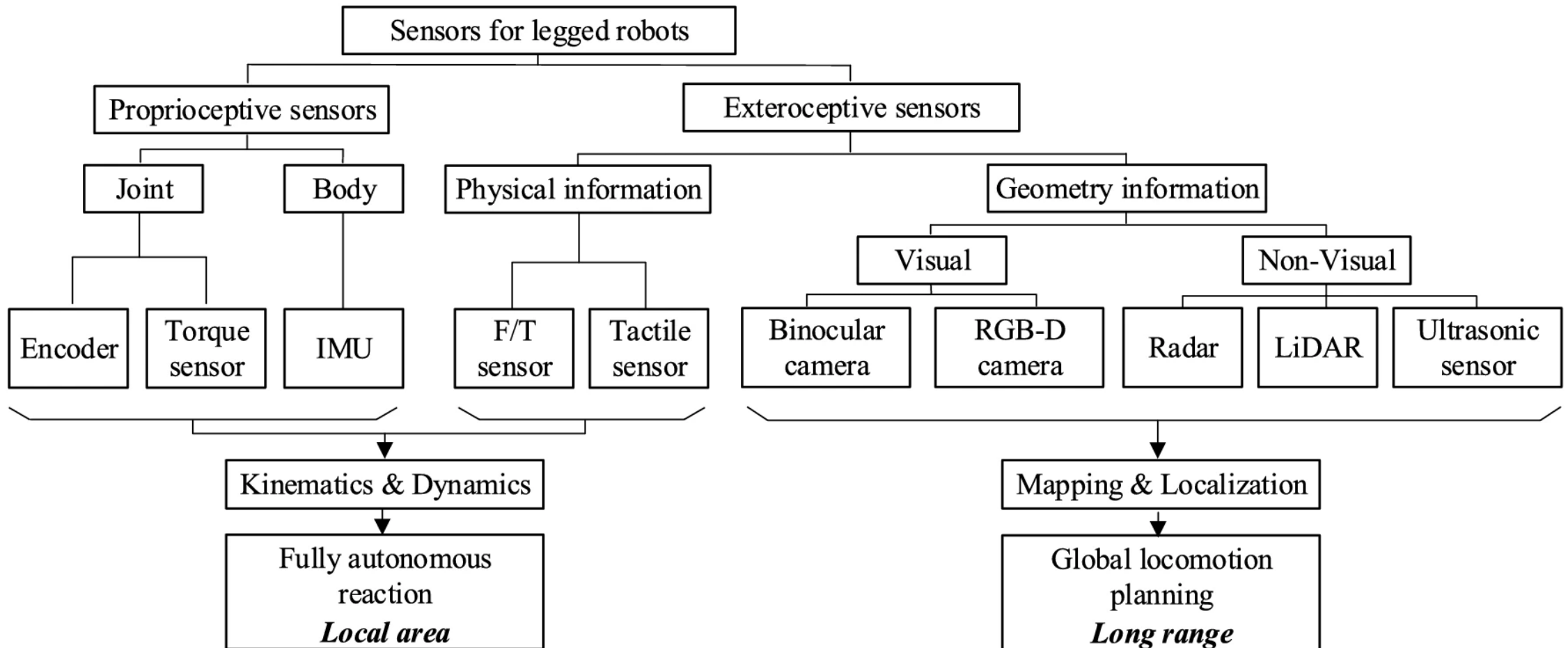
## Actuators & Sensors

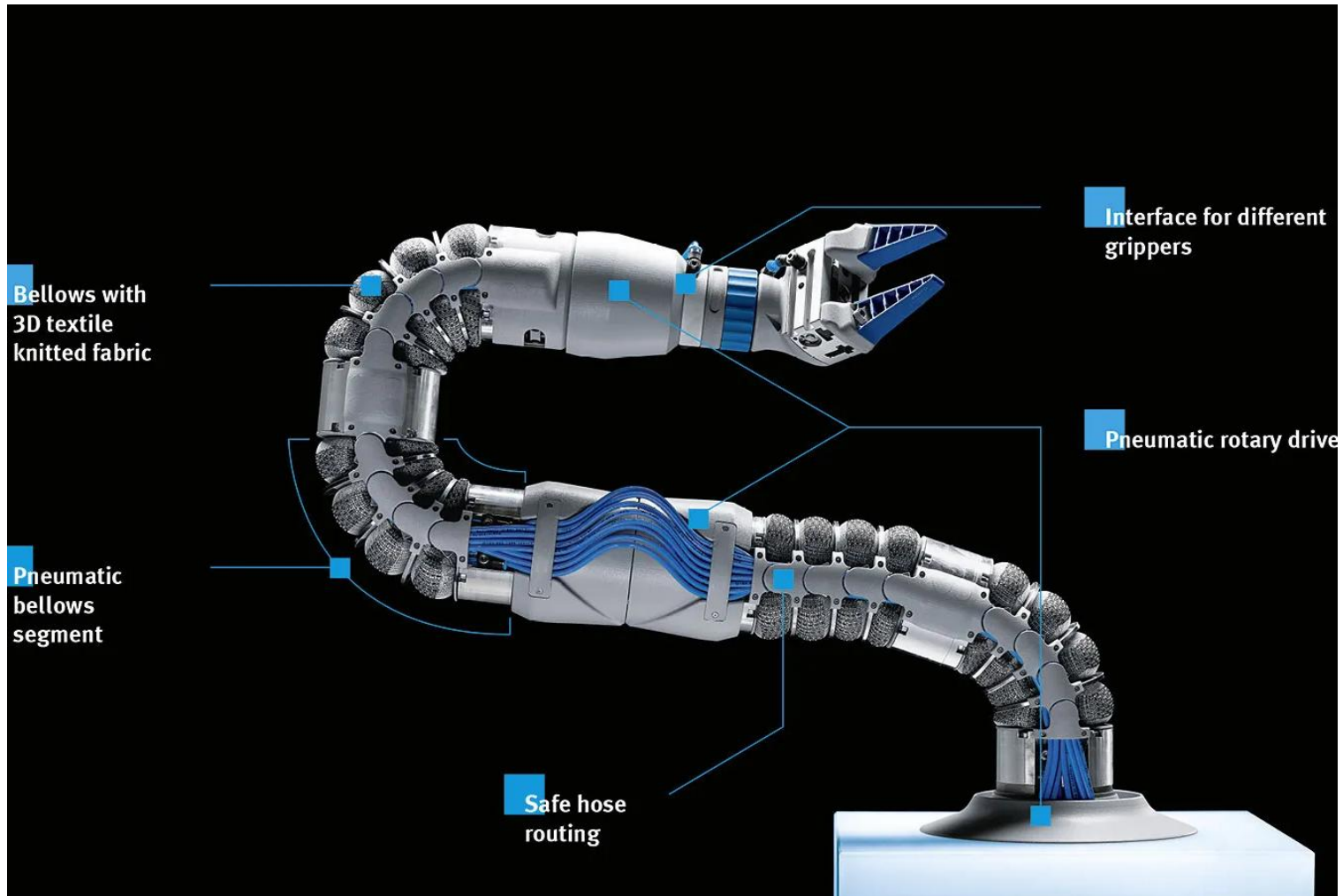


# Legged robots – the sensor & actuators

- <https://www.youtube.com/watch?v=rVlhMGQgDkY>
- <https://www.youtube.com/watch?v=bKDhmENcKto>
- <https://www.youtube.com/watch?v=7atZfX85nd4>

# Case Study – sensors in legged robots

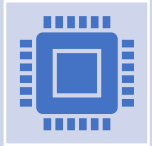




BionicSoftArm: Modular pneumatic lightweight robot



# Actuators



Devices that take signal in electrical form and transform it to something that can influence the physical world



Can almost say that this is the end goal of all IoT devices, i.e., influencing or altering the physical world



Can obtain actuation in many forms – movement, temperature (heating/cooling), light, sound, etc.

# Types of actuators

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To enact movement, the actuator requires energy. The main types of energy sources are the following:

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electric

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hydraulic

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pneumatic

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thermal/magnetic

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# Electric actuators

- **Advantages of electric actuators**

1. These actuators offer the highest precision among other actuators.
2. Easy to network and easy to program. They offer immediate feedback for diagnostic and maintenance.
3. They provide complete control on motion profiles and can include an encoder to control the velocity, position, and torque.
4. Less noise compared to hydraulic and pneumatic actuators
5. No fluid leak, so fewer environmental hazards.

- **Disadvantages of electric actuators**

1. The initial cost of the electrical actuator is higher
2. Unlike pneumatic and hydraulic actuators, these actuators are not suitable for all environments.
3. There are overheating, wear and tear issues there compared to pneumatic and hydraulic actuators.
4. The actuator's parameters are fixed, so to change torque, speed, etc to a different level, actuators should replace.

# Making things move

- First kind of actuation is electromagnetic drives
- Most common actuation technique used to get mechanical motion from electrical energy
- Motion can then be converted to displacement, force, pressure
- Workhorse of the industrial society





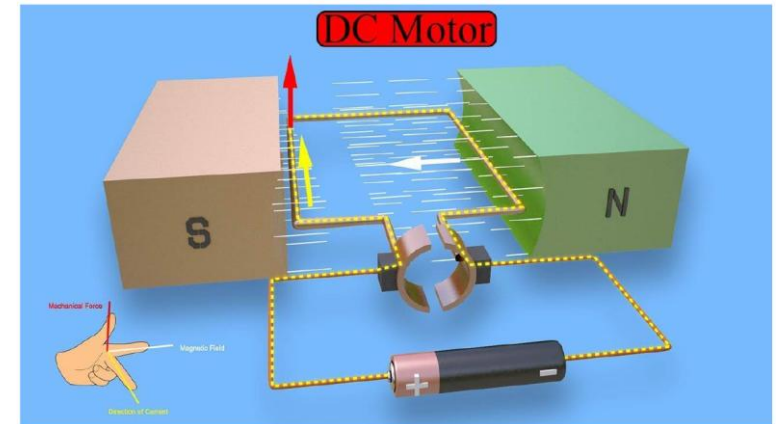
# Making things move

- Basic principle is very simple – uses two fundamental laws of physics:
  - When a current is flown through a conductor, a magnetic field is produced
  - Two magnets experience a force that tries to align their magnetic field
- Magnetic fields are typically produced by a stator (fixed) and rotor (movable)
- The fields can be produced using a permanent magnet or electromagnets



# DC Motor

- One of the simplest designs for electromagnetic drives
- Stator is a permanent magnet and rotor is a coil connected to the power supply
- When the current flows through the rotor coils, they become electromagnets and try to align with the stator field, causing rotation



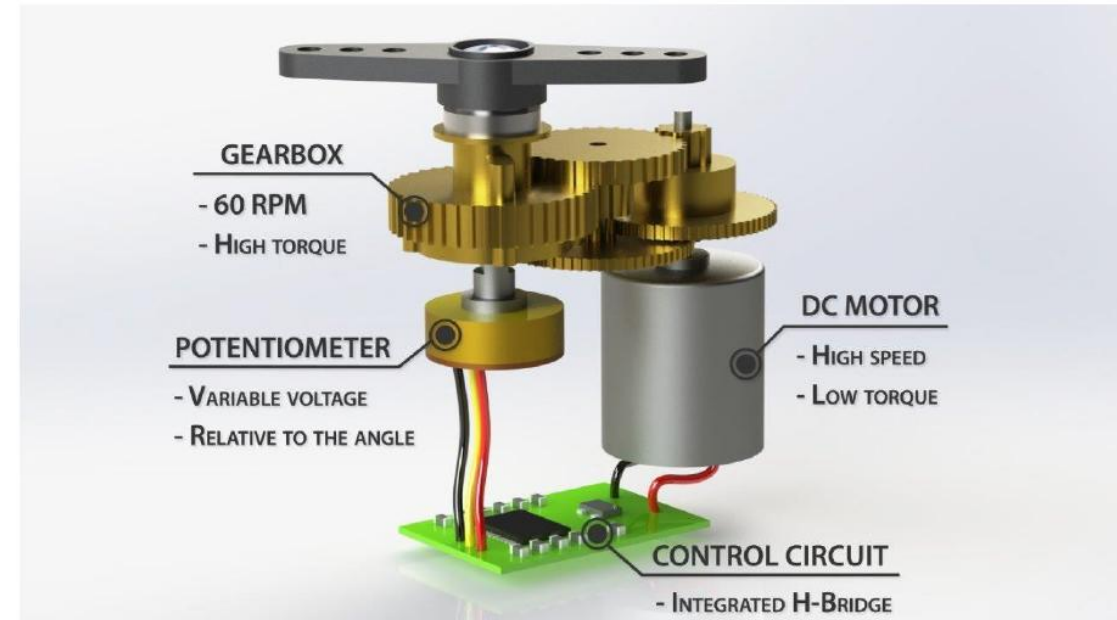
# Stepper Motor

- Class of motors used to produce rotational motion in exact increments
- Typically has a permanent magnet rotor that aligns to the magnetic field of the stator coils
- When the next stator coil is energized, the rotor moves one step and locks in place
- This mechanism yields precise movement



# Servo Motor

- Commonly used in robotics and industrial applications where a knowledge of the shaft position is required
- Uses a position sensor to control the motor (closed loop system)
- Control circuit provides feedback on the current position of the motor shaft → allows to rotate with great precision
- A gear assembly is used to obtain high torque



<https://www.youtube.com/watch?v=LXURLvga8bQ&t=27s>

**MG996R Servo Motor – good for most projects**

# Classification on type of displacement

- 1.Linear Actuators:** The shaft of the linear actuators will only move in a linear fashion.
- 2.Rotary actuators:** The shaft of the rotary actuator will only rotate in an axis.

# Hydraulic actuators

## Advantages of Hydraulic actuator

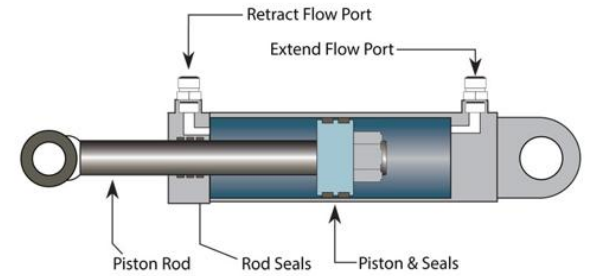
1. Easy to control and accurate
2. Simpler and easier to maintain
3. Constant torque or force regardless of speed changes
4. Easy to spot leakages of system
5. Less noise

## Disadvantages of Hydraulic actuator

1. Proper maintenance is required
2. Expensive
3. Leakage of the fluid creates environmental problems
4. Wrong hydraulic fluid for a system can damage the components



Hydraulic Actuator



Section of Hydraulic Actuator



# Pneumatic actuators

Similar to Hydraulic, use air instead of fluid

## Advantages of Pneumatic actuators

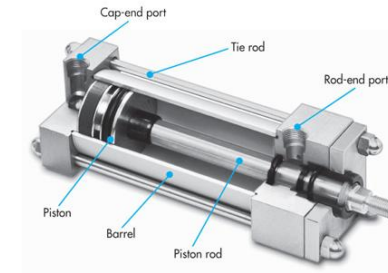
- 1.Clean, less pollution to the environment
- 2.Inexpensive
- 3.Safe and easy to operate

## Disadvantages of Pneumatic actuators

- 1.Loud and noisy
- 2.Lack of precision controls
- 3.Sensitive to vibrations



Pneumatic actuator



Section of pneumatic actuator

# Other examples

- Solenoid valves
- Electrical relays

# IoT/Actuators & Sensors in Automobile Industry

- <https://www.youtube.com/watch?v=bqU31EXPS-M>

# IoT – Robots

- Healthcare



The remote-controlled Da Vinci robot surgeon performs surgery

A doctor in Canada performs a surgery from 400km away

# Challenging Case

- Nizam Institute of Medical Sciences

OP : 9.2 lakh in 2024

Diagnostic tests: 34.6 lakh tests in 2024

Bed capacity: ~4000

ICU : ~1000

Employees : ~ 4000

Acres: 22

Stretchers: ~5000

Wheelchairs: ~ 5000

# IoT in Healthcare - robots

- [https://www.youtube.com/watch?v=pv24\\_19CVT4](https://www.youtube.com/watch?v=pv24_19CVT4)

